

SIMATS Online Programming Lab

PYTHON PROGRAMMING



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WAP to find smallest and largest number in an array

input: 12 5 18 3 25 9

output:

Smallest number:3

Largest number:25

Your Code

```
1 arr = list(map(int, input().split()))
2
3 small = arr[0]
4 large = arr[0]
5
6 for i in arr:
7     if i < small:
8         small = i
9     if i > large:
10        large = i
11
12 print("Smallest number:"+str(small))
13 print("Largest number:"+str(large))
```

Input

12 5 18 3 25 9

Output

Smallest number:3
Largest number:25



← Back

Q2 Missing Number

10 marks

WAP to find missing number in an array of shuffled order

input: 4 2 1 6 5

output:3

Your Code

```
1 arr = list(map(int, input().split()))
2
3 n = len(arr) + 1
4
5 total = n * (n + 1) // 2
6 arr_sum = sum(arr)
7
8 print(total - arr_sum)
```

Input

4 2 1 6 5

Output

3



Q3 Second largest number

10 marks

WAP to find second largest number in an array

INPUT: 12 5 18 3 25 9

OUTPUT: 18

Your Code

```
1 arr = list(map(int, input().split()))
2
3 first = second = float('-inf')
4
5 for i in arr:
6     if i > first:
7         second = first
8         first = i
9     elif i > second and i != first:
10         second = i
11
12 print(second)
```

Input

12 5 18 3 25 9

Output

18



Q4 Most repeated number

10 marks

WAP to print the most repeated number in an array

INPUT: 2 4 6 2 5 3 1 2

OUTPUT: 2

Your Code

```
1 arr = list(map(int, input().split()))
2
3 max_count = 0
4 most = arr[0]
5
6 for i in arr:
7     count = arr.count(i)
8     if count > max_count:
9         max_count = count
10        most = i
11
12 print(most)
```

Input

2 4 6 2 5 3 1 2

Output

2



← Back

Q5 Unique elements

10 marks

WAP to print all unique elements in an array

INPUT : 2 4 6 2 5 6 7

OUTPUT : 4 5 7

Your Code

```
1 arr = list(map(int, input().split()))
2
3 for i in arr:
4     if arr.count(i) == 1:
5         print(i, end=" ")
```

Input

2 4 6 2 5 6 7

Output

4 5 7



← Back

Q6 Rotate an array by 'N' positions

10 marks

WAP to rotate an array by 'N' positions

INPUT: 2 5 6 1 3 4

OUTPUT: 6 1 3 4 2 5

Your Code

```
1 arr = list(map(int, input().split()))
2 n = 2
3
4 arr = arr[n:] + arr[:n]
5
6 print(*arr)
```

Input

2 5 6 1 3 4

Output

6 1 3 4 2 5



← BACK

Q7 Remove all duplicates in an array

10 marks

WAP to remove all duplicates in an array

INPUT: 2 4 6 2 7 8 3 4

OUTPUT: 2 4 6 7 8 3

Your Code

```
1 arr = list(map(int, input().split()))
2
3 result = []
4
5 for i in arr:
6     if i not in result:
7         result.append(i)
8
9 print(*result)
```

Input

2 4 6 2 7 8 3 4

Output

2 4 6 7 8 3



Q8 Print all 0's followed by non-zero without changing the order

10 marks

WAP to print all 0's followed by non-zero without changing the order

INPUT: 1 0 4 0 2 5 0 7 0 6 0

OUTPUT: 0 0 0 0 0 1 4 2 5 7 6

Your Code

```
1 arr = list(map(int, input().split()))
2
3 zero = []
4 nonzero = []
5
6 for i in arr:
7     if i == 0:
8         zero.append(i)
9     else:
10        nonzero.append(i)
11
12 result = zero + nonzero
13
14 print(*result)
```

Input

1 0 2 0 4 5 0 7 0 6 0

Output

0 0 0 0 0 1 2 4 5 7 6

Visible Test Cases

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PYTHON PROGRAMMING



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Write a python program to count vowels, consonants, digits, and spaces in a given sentence

INPUT: Hello World 123

OUTPUT:

Vowels:3

Consonants:7

Digits:3

Spaces:2

Your Code

```
1 s = input()
2
3 vowels = consonants = digits = spaces = 0
4
5 for ch in s:
6     if ch in "aeiouAEIOU":
7         vowels += 1
8     elif ch.isalpha():
9         consonants += 1
10    elif ch.isdigit():
11        digits += 1
12    elif ch == " ":
13        spaces += 1
```

Input

Hello World 123

Output

```
Vowels: 3
Consonants: 7
Digits: 3
Spaces: 2
```



Q10 Palindrome

10 marks

Write a python program to check whether the given string is a palindrome

input: malayalam

output: malayalam is a palindrome

input: saveetha

output: saveetha is not a palindrome

Your Code

```
1 s = input()
2
3 if s == s[::-1]:
4     print(s, "is a palindrome")
5 else:
6     print(s, "is not a palindrome")
```

Input

malayalam

Output

malayalam is a palindrome